

BIOMECHANICS (EPHE 341)

Introduction and Review

What this subject is, how this course will do it, and the mathematics the rest of the term is going to assume you have.

What is biomechanics?

The study of the effects of **internal and external forces** on the human body in movement and at rest.

Bio — any biological system.

Mechanics — the branch of the physical sciences concerned with the state of rest or motion of bodies subjected to the action of forces.
Rigid body, fluid, deformable body.

The word to notice is **internal**. External forces can be measured — you stand on a plate and read the number. Internal ones cannot, and they are the ones that decide whether a tendon tears or a hip replacement lasts.

Almost everything this field knows about what the inside of a body is doing is *calculated*, not measured. The last three lectures of this course are how that calculation works.

Qualitative and quantitative

Qualitative — describing movement without numbers. A coach watching a stride, a clinician watching a gait. That is EPHE 201.

Quantitative — putting numbers on it. That is this course.

Neither is the better one. A clinician has fifteen minutes and needs a judgement; a researcher testing a new rehabilitation protocol needs a measurement. Each answers questions the other cannot.

What the quantitative approach buys is the ability to be **wrong in a checkable way** — which is the whole reason this course spends so much time on units and sanity checks.

The four words you will hear all term

Statics — bodies at rest, or moving at constant velocity, with balanced forces acting. *Lecture 19.*

Dynamics — the relationship between motion and the forces that cause it. *Lecture 20.*

Kinematics — the spatial and temporal components of motion: position, velocity, acceleration, with no regard to force. The *description* of movement.

Kinetics — bodies in motion with respect to mass, force and energy. The *causes* of movement.

Kinesis, from the Ancient Greek, is motion — and *kineo* is to put in motion. Kinematics describes; kinetics explains. Half the course is each.

What it is for

To measure or predict internal and external forces, in:

- **occupational and industrial** settings — ergonomics, and designing a task so it injures fewer people;
- **clinical** settings — assessing whether a rehabilitation method works;
- **sport** — assessing a technique, and finding out whether a change to it helped.

And the measurement tools themselves are part of the subject: force plates, three-dimensional motion capture, electromyography, accelerometers. You will use most of them.

Nearly every figure in this course and the nineteen that follow it is driven by a **real recording** rather than a drawing. Where one is a model rather than a measurement, the figure says so on its face.

The whole course, on one slide

Force forces	Moment of Force / Torque angular kinetics 1	Static Equilibrium static analysis	Friction linear kinetics 1	Linear Kinematics calculus and kinematics	Angular Kinematics angular kinematics	General Kinematics general kinematics
Basic Calculus calculus and kinematics ↗	Projectile motion projectile motion	Anthropometry angular kinetics 2	Moment of Inertia angular kinetics 3	Dynamic Equilibrium dynamic analysis	Work / Energy work energy power	Power work energy power ↗
Impulse Momentum linear kinetics 2	3D Forces and Moments gait analysis	Current measurement tools sensors and data acquisition	Force measurement forces ↗	3D motion analysis gait analysis ↗	Electromyography electromyography	Mechanical properties of muscle muscle mechanics
Gait analysis gait analysis ↗	Signals signals					

His slide 11, with the lectures attached. Every topic on that list is now an interactive deck — **18 of them** — and this is the only slide in the course that can say so. That is the argument for converting the introduction last: a map is not much use until there is something to map.

How this course works

A text you do not have to buy, problem sets that are not graded, and a tutor that is open all night.

How This Course Works

The text, the problem sets, the Tutor

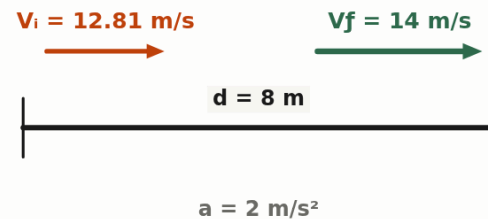
The text is optional. All test content comes from the lecture notes. *Biomechanics of Sport and Exercise* (McGinnis) supplements them and is good on qualitative analysis, which this course does not cover in depth.

The problem sets are not graded — and the test material is based on them. Those two sentences are not in tension: the problem sets are the practice, the test is the performance.

The Biomechanics Tutor works by leading questions: it will not just mark your final answer, it walks you through the step you got stuck on.

One warning you will be glad of. The Tutor grades your number against an exact range with **no tolerance**, so enter the rounded value it is expecting, not every digit your calculator produced.

How to solve a problem



step 11

$V_i = 12.81 \text{ m/s}$. $\sqrt{(\text{m}^2/\text{s}^2)}$ is m/s , so the units come out right too.

- 1 Read the problem carefully to understand what is asked.
- 2 Write down what is to be determined.
- 3 When helpful, draw a diagram.
- 4 Identify and write down helpful formulas related to the variable.
- 5 Write down all the given information relevant to those formulas.
- 6 Select the formula that contains the givens and the unknown.
- 7 If no workable formula can be found, return to step 1.
- 8 Write it out and substitute the known quantities.
- 9 Use basic algebra to solve for the unknown.
- 10 Check to see if the answer makes sense.
- 11 Box the answer and check the units.

His slide 16, one step at a time, on the Tutor's first Basic Math question.

Step 11 of 11

Eleven steps, and the two that matter most are the ones that look like filler. **Step 6** is where the problem is actually solved: two of the three kinematics equations need a time nobody gave you, so there is only one left and no choice to agonise over. **Step 10** is the one that catches mistakes — including the one on his own slide 38.

Before next class

Most of what follows should have been learned in high-school mathematics. It is **crucial**, and it is your responsibility to be able to manipulate numbers and equations confidently.

- basic maths
- trigonometry
- basic functional anatomy
- linear kinematics

Homework. Review the rest of this deck, work the problem sets for basic maths and trigonometry, and review the linear kinematics lecture.

The students who find this course hard are almost never the ones who find *mechanics* hard. They are the ones who cannot rearrange an equation at speed. That is fixable, and now is when it is cheap to fix.

The mathematics

Real numbers, exponents, and three kinds of equation. Nothing here is new; all of it has to be fast.

Three properties, and everything else

Commutative — order does not matter.

$$a + b = b + a \quad a \times b = b \times a$$

Associative — grouping does not matter.

$$(a + b) + c = a + (b + c)$$

Distributive — a multiplier outside a bracket reaches every term inside it.

$$a(b + c) = a \cdot b + a \cdot c$$

Every one of the sixteen theorems on his slide 24, and every step of every rearrangement you will do this term, is one of these three applied once.

Twenty-four identities, all of them true

1 $c^n \cdot c^m = c^{n+m}$

$2^3 \times 2^5 = 2^{3+5} = 2^8$

✓ 256 = 256

2 $x^0 = 1$

$4^0 = 1$

✓ 1 = 1

3 $x^{-n} = 1/x^n$

$3^{-2} = 1/3^2$

✓ 0.125 = 0.125

4 $c^m / c^n = c^{m-n}$

$2^7 / 2^4 = 2^3$

✓ 0.25 = 0.25

5 $(c^n)^m = c^{nm}$

$(5^2)^3 = 5^6$

✓ 32768 = 32768

6 $(c \cdot d)^n = c^n \cdot d^n$

$(2 \times 3)^4 = 2^4 \times 3^4$

✓ 216 = 216

7 $(c/d)^n = c^n / d^n$

$(3/4)^5 = 3^5 / 4^5$

✓ 0.125 = 0.125

8 $x^{m/n} = \sqrt[n]{x^m}$

$3^{4/2} = \sqrt{(3^4)}$

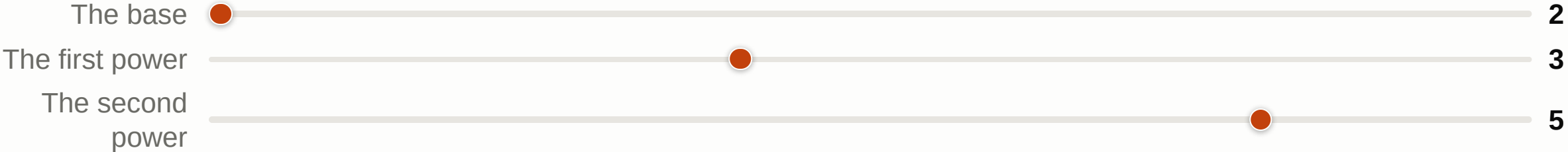
✓ 1.515717 = 1.515717

all eight hold, with his numbers and with these

The sliders change the base and the two powers · law 2 has no power to change and law 8 needs a whole root, so those two ignore them

The eight laws of exponents

The sixteen theorems



Twenty-four identities, and every one of them holds — with his numbers and with any others you slide to. That is worth saying out loud: his review content is **right**, and the only two things this deck corrects in fifty-nine slides are a quadratic’s roots and a cosine-law answer.

Three kinds of equation

One unknown. Isolate it. Whatever you do to one side you do to the other.

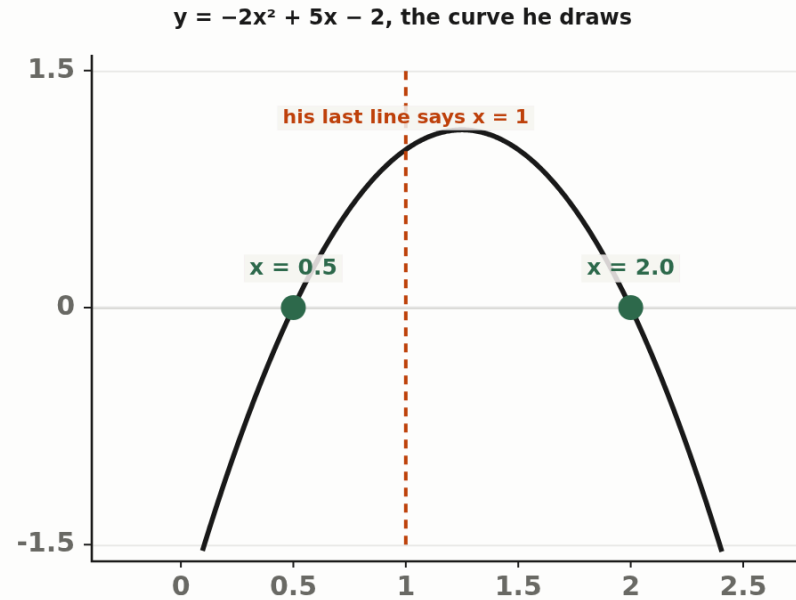
Two unknowns. You need two equations. Isolate a variable in one, substitute it into the other, solve, then back-substitute.

A quadratic. Anything of the form $ax^2 + bx + c = 0$, which has **two** answers.

$$x = (-b \pm \sqrt{b^2 - 4ac}) / 2a$$

Quadratics turn up in this course the moment anything is thrown: a projectile is at a given height **twice**, once going up and once coming down, and the two roots are those two times.

His three examples, with the answer drawn



$$-2x^2 + 5x - 2 = 0, \text{ find } x$$

the quadratic formula

$$x = (-b \pm \sqrt{b^2 - 4ac}) / 2a$$

substitute

$$x = (-5 \pm \sqrt{25 - 16}) / -4$$

so

$$x = (-5 \pm 3) / -4 = 0.5 \text{ or } 2$$

$$x = 0.5 \text{ and } x = 2.0$$

his slide concludes $x = \frac{1}{2}$ and $x = 1$

The formula on his slide is right and he applies it right. Only the concluding line is wrong — and his own graph, printed beside it, cuts the axis at $\frac{1}{2}$ and 2.
Reading the answer off the picture would have caught it, which is his own step eleven: check to see if the answer makes sense.

One unknown

Two unknowns

A quadratic

Three equations, three shapes of answer: a point on a line, a crossing of two lines, and the two places a curve cuts the axis. Drawing the answer is not decoration — it is the check. His slide 30 applies the quadratic formula correctly and then writes down the wrong pair of roots, and the graph printed on the same slide disagrees with the sentence underneath it.

Trigonometry

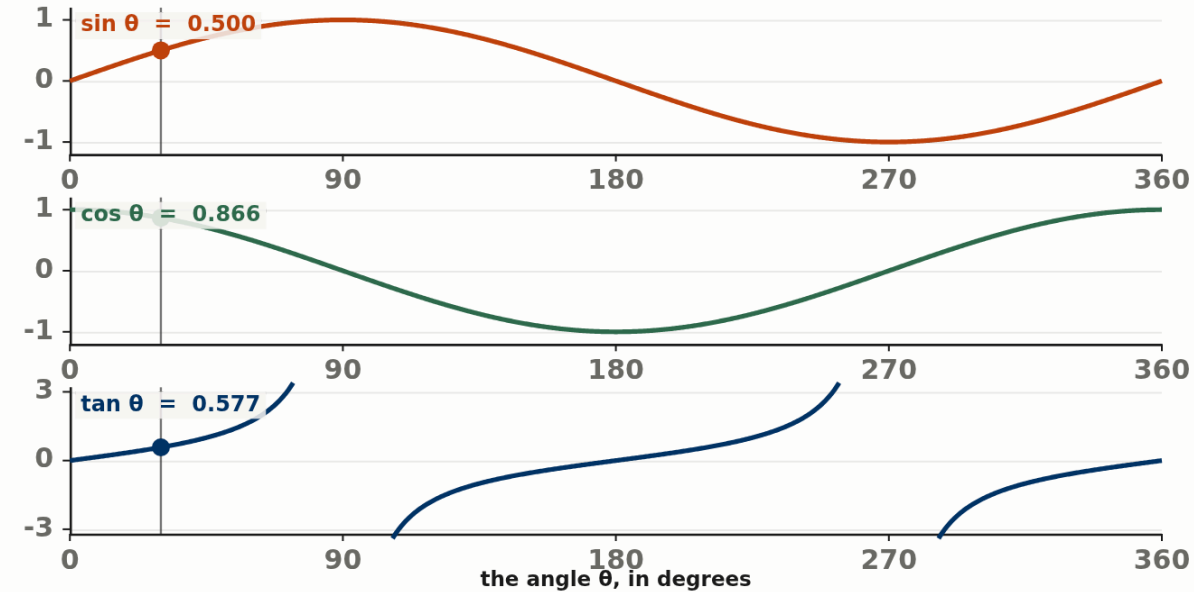
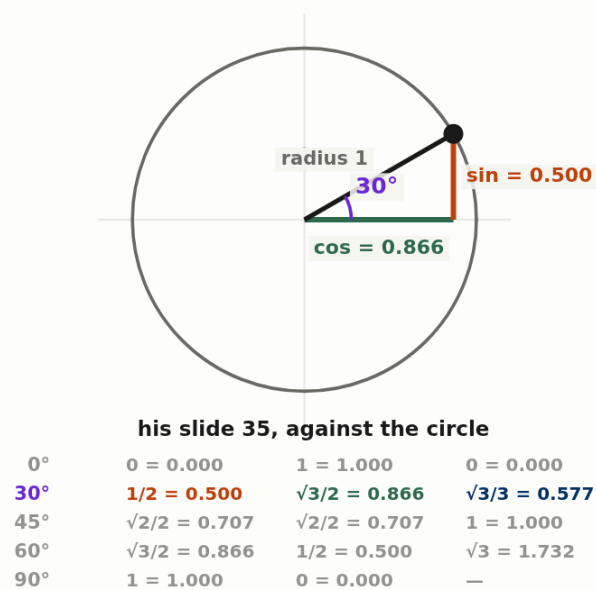
The part of the review that you will use every single week, because almost every force in this course arrives at an angle.

Eight things about lines and triangles

- Two lines are **perpendicular** when they meet at 90° .
- **Parallel** lines never intersect.
- Where two lines cross, **opposing angles are equal**.
- The angles around an intersection sum to **360°** .
- The angles along one line at an intersection sum to **180°** .
- A line crossing two parallels makes a **Z**, and the corresponding angles are equal.
- Three non-parallel lines make a **triangle**.
- A triangle's interior angles sum to **180°** .

The Z rule and the 180° rule between them are how you find an angle you were never given — which is most of what free body diagrams ask you to do.

Where sine and cosine come from



One point, one circle · its height is the sine, its sideways distance the cosine, and the slope of the radius the tangent · which is why tan blows up at 90°, where the radius is vertical and has no sideways distance left to divide by

The angle θ



0°

30°

45°

60°

90°

► Go round

Sine, cosine and tangent are not three functions to memorise, they are **one point going round one circle**. The height of the point is the sine, its sideways distance is the cosine, and the tangent is one divided by the other. Press the special angles and watch his slide 35 light up a row at a time — those five rows are five places on this circle, which is a great deal easier than learning a table.

Three relations, and when to use each

Right-angle triangle.

$$\sin\theta = \text{opposite} / \text{hypotenuse}$$

$$\cos\theta = \text{adjacent} / \text{hypotenuse}$$

$$\tan\theta = \text{opposite} / \text{adjacent}$$

and Pythagoras: $\text{hypotenuse}^2 = \text{adjacent}^2 + \text{opposite}^2$.

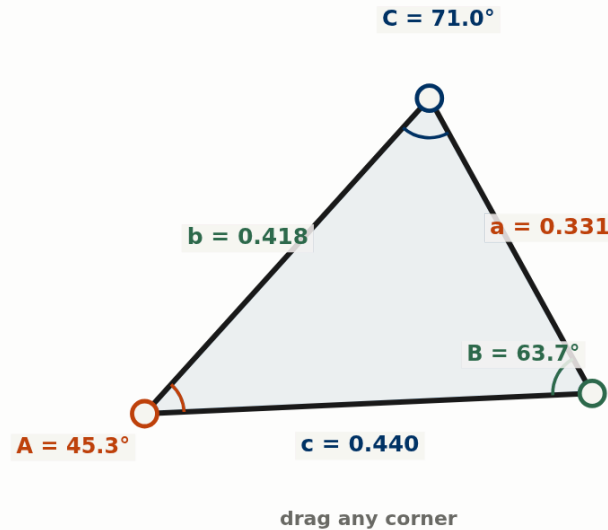
Any triangle.

$$a / \sin A = b / \sin B = c / \sin C$$

$$a^2 = b^2 + c^2 - 2bc \cdot \cos A$$

Which one you use is decided by **what you were given**, not by what you want. Two angles and a side → the sine law. Two sides and the angle between them → the cosine law.

One triangle, all three relations at once



the angles always sum to 180°

$$45.3^\circ + 63.7^\circ + 71.0^\circ = 180.0^\circ$$

the sine law — $a/\sin A = b/\sin B = c/\sin C$

$$0.466 \quad 0.466 \quad 0.466 \quad (\text{all one number})$$

the cosine law — $c^2 = a^2 + b^2 - 2ab \cdot \cos C$

$$0.331^2 + 0.418^2 - 2(0.331)(0.418)\cos 71.0^\circ \rightarrow c = 0.440, \text{ and } c \text{ measures } 0.440$$

Which law you use is decided by what you were given, not by what you want: two angles and a side → the sine law; two sides and the angle between them → the cosine law; a right angle → either, or sin, cos and tan.

The sine and cosine laws

The right-angle functions

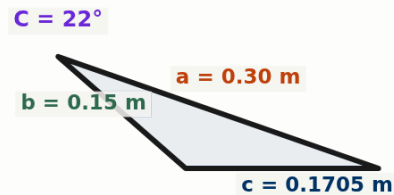
Make it right-angled

Make it equilateral

One triangle, three laws, all true at once. Drag a corner: the three sine-law ratios stay equal to each other while every one of them changes, and the cosine law reproduces the side you can measure. **Which law you reach for is decided by what you were given** — two angles and a side, the sine law; two sides and the angle between them, the cosine law.

His two worked examples

Example 2: a non-right-angle triangle



his 0.3227 m, to the same scale

What is the length of side c , given angle $C = 22^\circ$, side $a = 0.30 \text{ m}$ and side $b = 0.15 \text{ m}$?

two sides and the angle between them, so this is the cosine law

$$c^2 = a^2 + b^2 - 2ab \cdot \cos C$$

$$c^2 = 0.30^2 + 0.15^2 - 2(0.30)(0.15)\cos 22^\circ$$

$$c^2 = 0.1125 - 0.0834 = 0.029053$$

$$c = 0.1705 \text{ m}$$

his slide prints 0.3227 m

Look at the picture rather than the arithmetic. Drawn to his own givens, c comes out clearly shorter than the 0.30 m side — and the sketch on his slide 38 shows exactly that. The dashed line is what 0.3227 m would measure; it is longer than every side of the triangle it is supposed to close.

His Example 1

His Example 2

The Tutor's Q13

His Example 1 reproduces to the digit. **Example 2 does not:** the cosine law with his own three givens gives **0.1705 m** where the slide prints 0.3227 m. The useful part is not the correction, it is the check — his own eleventh step of problem solving is "check to see if the answer makes sense", and a side of 0.32 m cannot close a triangle whose other two sides are 0.30 and 0.15 with only 22° between them. Drawing it to scale catches the error in one second.

Where things are

Three planes, three axes, and a vocabulary for saying which way a limb went.

The three planes

Sagittal — divides the body into left and right. Most of walking happens here, which is why most of this course is two-dimensional.

Frontal — divides front from back.

Transverse — divides top from bottom.

An **axis** is the straight line where two planes intersect — and it is also the line a body rotates about.

Those two sentences describe the same line, and that is not a coincidence. A joint that moves in one plane rotates about the axis perpendicular to it. Flexing your knee is movement in the sagittal plane about a left–right axis.

And the three axes they make

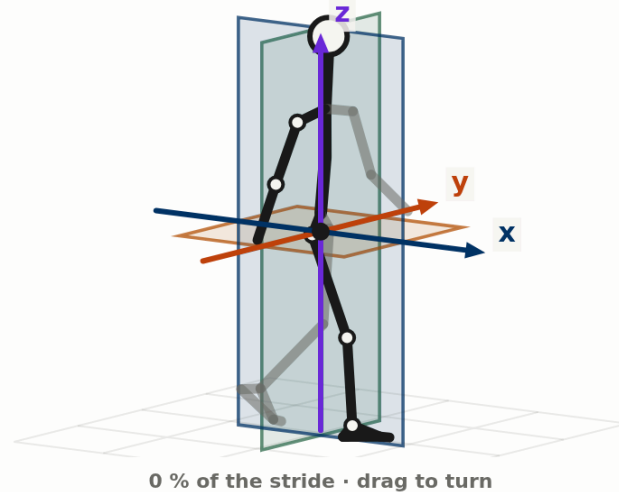
- sagittal ∩ frontal = the **z** axis — distance along it is up and down;
- frontal ∩ transverse = the **y** axis — left and right;
- sagittal ∩ transverse = the **x** axis — front and back.

That is the convention every figure in this course uses, and it is the convention the motion-capture system in the lab writes its files in. When you open a data file in week ten, these are the columns.

Every number you will ever read off a marker, a force plate or an accelerometer is a distance or a force *along one of these three lines*.

Planes and Axes

On somebody who is actually moving



A measured walking stride · the planes are fixed in the room and the body moves through them, which is his slides 47 to 50: the joint axes go with the limbs, not with the diagram

his slides 39 to 46, in one figure

- **the sagittal plane**
divides left from right
- **the frontal plane**
divides front from back
- **the transverse plane**
divides top from bottom
- **the z axis**
where the sagittal and frontal planes meet
distance along it is a measure of up and down
- **the y axis**
where the frontal and transverse planes meet
distance along it is a measure of left and right
- **the x axis**
where the sagittal and transverse planes meet
distance along it is a measure of front and back

Where in the stride 0 %

Everything

Sagittal

Frontal

Transverse

z

y

x

▶ Walk

Three planes, three axes, and a real person walking through them. The convention is his, off slides 41 to 46: **z** where sagittal and frontal meet, up and down; **y** where frontal and transverse meet, left and right; **x** where sagittal and transverse meet, front and back. Drag to turn it; press Walk and the planes stay put while the limbs swing out of them.

Practice

Twenty-nine questions you can do tonight, and one of them is broken.

The Tutor, Basic Math and Trigonometry

Basic Math, question 1

Solve for V_i in $V_f^2 = V_i^2 + 2ad$, given $V_f = 14$ m/s, $a = 2$ m/s², $d = 8$ m

$$V_f^2 = V_i^2 + 2ad, \text{ so } V_i = \sqrt{V_f^2 - 2ad}$$

$$\sqrt{(14^2 - 2(2)(8))}$$

$$= 12.8062 \text{ m/s}$$

the Tutor accepts only 12.81 m/s

— which is this answer to 2 decimals, so it is right

The Tutor grades with a bare $\min \leq \text{answer} \leq \max$ and no tolerance at all, so a band with both ends the same is an exact match on the digits — enter the rounded value it shows, not the full one.

Basic Math

Trigonometry

1

2

3

4

5

6

7

8

9

10

11

12

Twenty-nine questions and twenty-eight of them are exactly right. **Basic Math question 6 is not:** its own printed method, $V_i = (d - \frac{1}{2}at^2)/t$, gives **19 m/s** from its own givens, and the answer it accepts is 5.5 — what that formula gives for $d = 46$ m rather than 100. Do it correctly and you are marked wrong. **Trigonometry question 15** has a milder version of the same thing: it wants 11.27° where the full calculation gives 11.28° . Both are worth reporting, and worth knowing before you lose ten minutes to either.

What to take away

- Biomechanics measures the **outside** of a body and computes the **inside**.
- **Kinematics** describes motion; **kinetics** explains it. You need both.
- Which trigonometric law you use is decided by **what you were given**.
- An **axis** is where two planes meet, and it is the line a joint turns about.
- The algebra is not the hard part of this course, but it is the part that costs marks. Fix it now.
- **Draw the answer.** Both of the errors in this lecture would have been caught by one sketch.

Next

Next class is **sensors and data acquisition** — how a force becomes a number, and what happens to it on the way. After that, calculus and kinematics, and then forces, and from there the course is continuous to the end.

Everything on the map earlier in this deck is live, works on a phone, and has a handout. The figures with sliders are the ones worth half an hour the night before a test.

Work the basic maths and trigonometry problem sets before next class.